

Splunk Fundamentals

Data to Everything

Monitoring, Searching, and Visualizing Data



60-Minute Training Agenda

01

Introduction to Splunk

Platform overview and core concepts

03

Data Ingestion & Sources

Methods for collecting and importing data

05

Dashboards & Alerts

Visualization and monitoring tools

02

Architecture & Components

System design and deployment modes

04

Search Processing Language

SPL basics and query examples

06

Hands-on Examples

Practical demonstrations and Q&A

What is Splunk?

Splunk is a comprehensive platform for searching, monitoring, and analyzing machine data. It works seamlessly with structured, semi-structured, and unstructured data, making it invaluable for organizations dealing with massive amounts of information.

The platform transforms raw data into actionable insights, enabling businesses to make informed decisions based on real-time analytics and historical trends.

IT Operations

Log analysis and troubleshooting

Security (SIEM)

Threat detection and incident response

Business Analytics

Customer behavior and transactions



Real-World Use Case

Banking Security Example

A major bank uses Splunk to detect failed login attempts and alert security teams in real time. When suspicious activity is detected, automated alerts trigger immediate investigation protocols, preventing potential security breaches.

This example demonstrates how Splunk transforms raw security logs into actionable intelligence, enabling proactive threat detection and rapid response capabilities across enterprise environments.



Splunk Architecture Components



Forwarder

Collects logs and sends data to indexer for processing



Indexer

Stores and indexes data for fast search and retrieval



Search Head

User interface for searching and visualization



Management

Deployment Server and Cluster Master for scaling

Deployment Modes

Standalone

Perfect for small setups and testing environments

Distributed

Separate indexers and search heads for better performance

Clustered

High availability with redundancy and load balancing

Data Ingestion & Sources



Logs

Server logs, application logs, and system events provide detailed operational insights



Metrics

System performance data, cloud monitoring metrics, and KPI measurements



Events

Security events, audit trails, and compliance-related data streams

Ingestion Methods

Universal Forwarder (UF)

Lightweight agent for continuous data collection

HTTP Event Collector

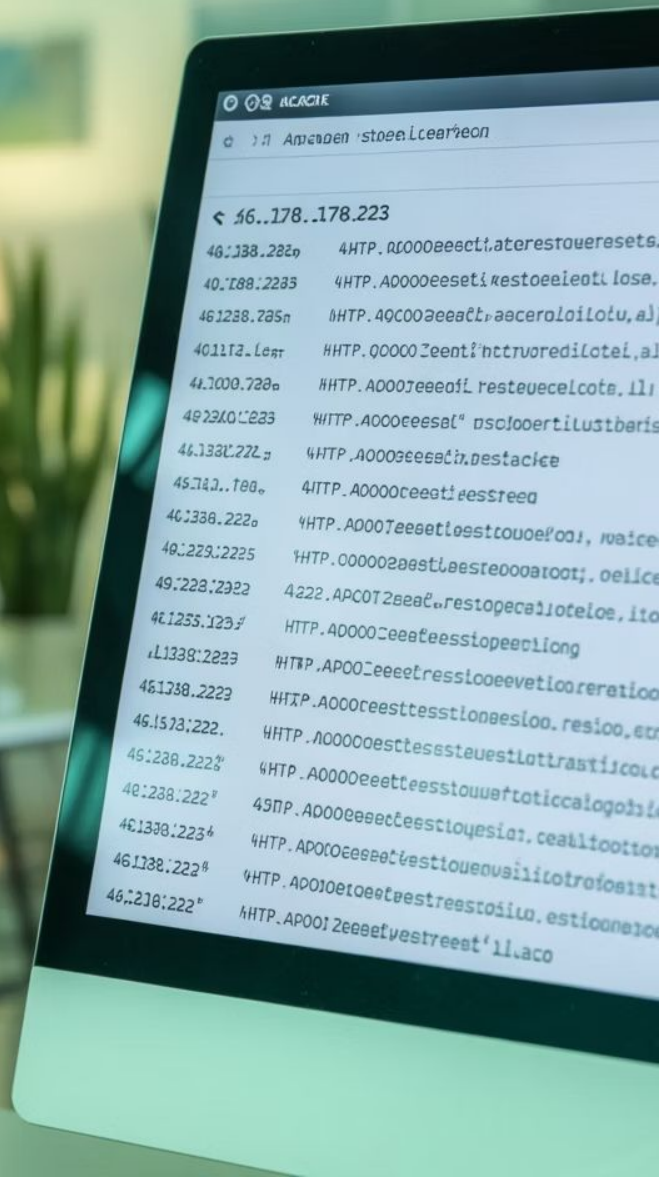
REST API for direct data submission

Manual Upload

CSV/JSON file imports for batch processing

API Integrations

Third-party system connections



Apache Web Server Log Example

When uploading Apache Web Server logs into Splunk, the platform automatically extracts valuable fields including IP addresses, requested URLs, response codes, and timestamps. This structured approach enables rapid analysis of web traffic patterns and performance metrics.



Key Fields Extracted: IP Address, URL, Response Code, Timestamp, User Agent, Referrer

Search Processing Language (SPL)

SPL is Splunk's powerful query language for searching and analyzing data. Similar to SQL but optimized specifically for logs and events, SPL enables complex data analysis with intuitive commands.



search

Filter events based on criteria



stats

Aggregate and summarize data



eval

Create new calculated fields



timechart

Time-series visualization



where

Apply conditional filters

Sample SPL Queries

```
index=security sourcetype=linux_secure "Failed password"
```

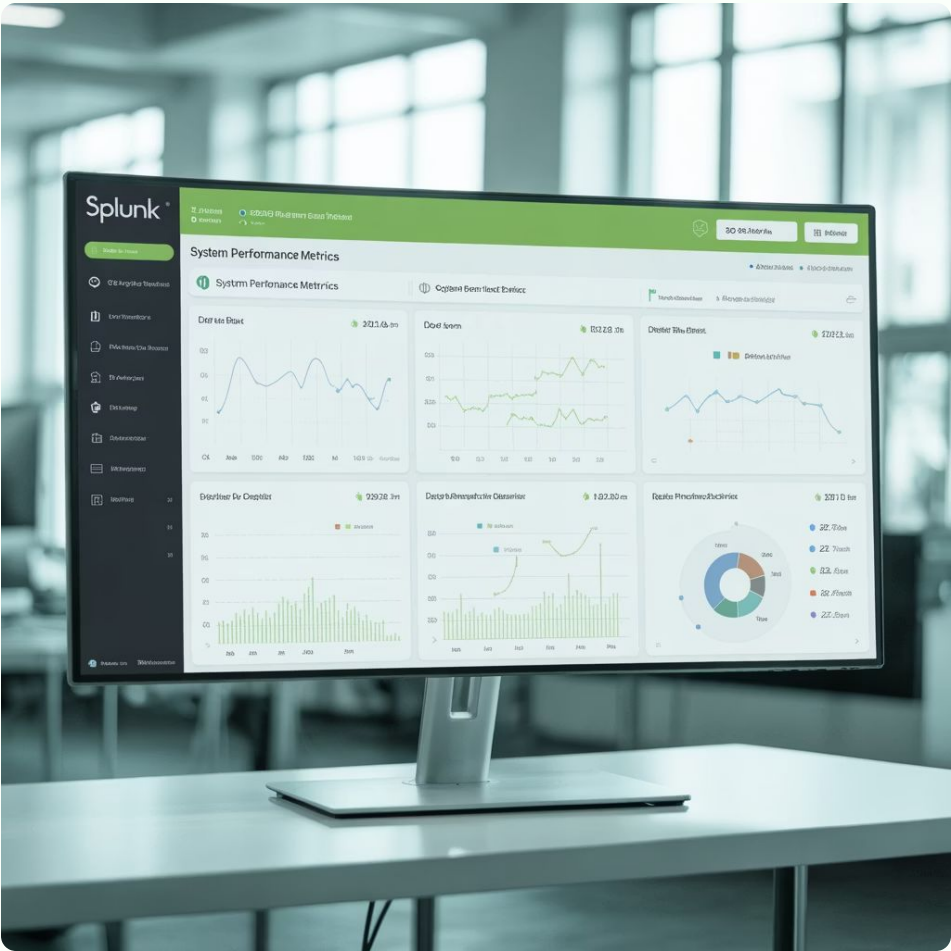
```
index=syslog | stats count by host
```

```
index=web sourcetype=access_combined | timechart count by status
```


Dashboards, Reports & Alerts

Dashboards

Combine multiple visualizations into comprehensive monitoring views. Real-time data updates provide instant insights into system performance and business metrics.



Alerts & Reports

Automated notifications triggered by specific conditions. Reports can be scheduled and exported to various formats for stakeholder distribution.

Email Notifications

Instant alerts via email

Slack Integration

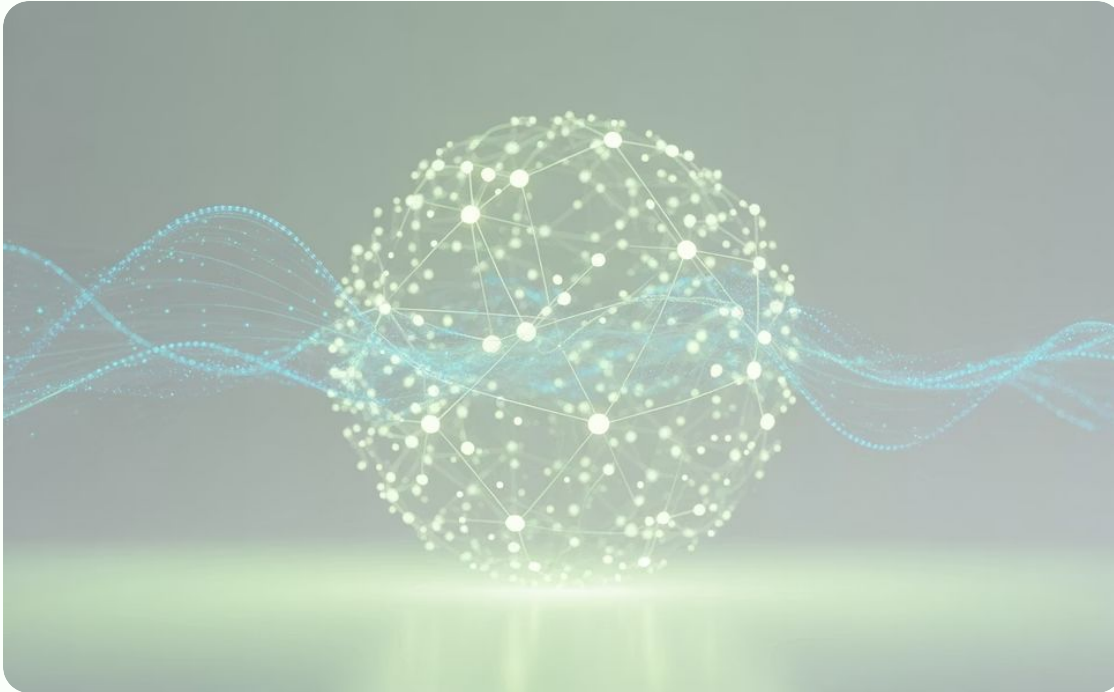
Team collaboration alerts

Webhook Support

Custom system integrations

⚠ **Example Alert:** Trigger when failed logins exceed 5 attempts within 1 minute - Dashboard showing top 10 IPs with failed login attempts

Advanced Features



Machine Learning Toolkit

Predict anomalies in logs and detect unusual patterns. Example: automatically identify abnormal CPU usage spikes that could indicate system issues or security threats.



Splunk Apps & Add-ons

Pre-built integrations for AWS, Azure, Kubernetes, and hundreds of other platforms. These apps provide ready-to-use dashboards and reports tailored for specific technologies.

Training Summary & Time Allocation

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Sources and methods

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SPL Basics

Query language fundamentals

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Dashboards

Visualization and alerts

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Advanced & Q&A

ML features and discussion

Key Takeaways

- Splunk transforms raw data into actionable insights through powerful search and analytics capabilities
- SPL is the core language for searching, filtering, and reporting on machine data
- Dashboards and alerts enable real-time monitoring and proactive incident response
- Advanced features like machine learning and pre-built apps extend platform capabilities



Questions & Next Steps

Q&A

Ready to explore your specific use cases and implementation questions

Ready-to-Use Content

All slides, diagrams, and sample queries formatted for immediate presentation use

Hands-On Practice

Try the SPL examples with your own data sources and explore dashboard creation

Implementation Support

Contact for assistance with specific deployment scenarios and advanced configurations